Diophantine Equations Problem Set 1

Md. Fuad Al Alam

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Problem 1. (Rioplatense MO 2023 P1) Find all positive integers (x, y, p) such that p is prime, $p = x^2 + 1$ and $2p^2 = y^2 + 1$.

Problem 2. (Pan American Girls' Mathematical Olympiad 2022 P2) Find all ordered triplets (p, q, r) of positive integers such that p and q are two (not necessarily distinct) primes, r is even, and

$$p^3 + q^2 = 4r^2 + 45r + 103$$

Problem 3. (Kazakhstan Regional 2021 Grade 9 P3) Find all pairs (x, y) of positive integers that satisfy the equation

$$125 \cdot 2^x - 3^y = 271$$

Problem 4. (Belarus MO 2024 Grade 10 P6) Find all pairs (p, q) of prime numbers satisfying the equation

$$p^{2q+1} = q^p + 2023$$

Problem 5. (Al-Khwarizmi IJMO 2025 P1) Determine the largest integer c for which there exists at least one triple (x, y, z) of integers such that

$$x^2 + 4(y + z) = y^2 + 4(z + x) = z^2 + 4(x + y) = c$$

and all triples (x, y, z) of real numbers satisfying the equations are such that x, y, z are integers.

Problem 6. (USAJMO 2023 P1) Find all triples of positive integers (x, y, z) that satisfy the equation

$$2(x + y + z + 2xyz)^2 = (2xy + 2yz + 2zx + 1)^2 + 2023$$

Problem 7. (Mongolia 2024 P1) Find all triples (a, b, c) of positive integers such that $a \leq b$ and

$$a! + b! = c^4 + 2024$$

Problem 8. (IMO 2022 P5) Find all triples (a, b, p) of positive integers with p prime and

$$a^p = b! + p$$